Forces and pressure

Moment: Turning effect of a force.

Moment = Force * Perpendicular distance from the point

$M = F \times d \rightarrow m$

$M \rightarrow Nm$, $F \rightarrow N$

A moment can be clockwise or anticlockwise.

To increase the moment:
- Increase the distance
- Increase the force

Principle of moment: For an object to be in equilibrium:
1- The sum of forces in one direction must be equal to the sum of forces in the opposite direction.
2- Clockwise moment = Anticlockwise moment

* If a force is passing through the pivot, it has zero moment
* The closer the force is to the pivot, the greater the value of that force.

Center of mass / gravity: A point in an object where the weight is considered to act.
* If an object doesn't have a constant thickness; is non uniform, the weight will act on the heavier side of the object.

To increase stability (make it harder to fall):
- Lower the center of mass
- Increasing the width of the base.

Extension: The length added to the original length.
x = Total new length - Original length.

* The extension is directly proportional to the load acting on the spring as long as the elastic limit is not crossed.

Hooke's law: $F \propto x$

Force $\propto$ Extension.

$F = Kx$
- F → force (N)
- x → extension (m)
- K → spring constant (N/M)

For a large value of K:
- Increase the Force
- Decrease Extension. ($K \propto \frac{1}{x}$)

Pressure: $\frac{Force}{Area}$

$P = \frac{F}{a}$
- $F \rightarrow N$
- $a \rightarrow m^2$
- P → Pa : N/m^2

$P \propto F$

$P \propto \frac{1}{a}$

1- Pressure acts in all directions
2- Pressure increases as depth increases
3- Pressure increases as density of the liquid increases
4- Pressure doesn't depend on the shape of the container

Pressure in liquid = $\rho g h$
- ρ → density
- g → gravity
- h → depth

Gas pressure & volume
1- Pressure $\propto$ temperature
2- Volume $\propto$ temperature
3- Pressure $\propto \frac{1}{Volume}$

$P_1 V_1 = P_2 V_2$